Advanced Calculus I Non-hand-in Assignment 3

This homework comes from textbook and will **NOT** be collected (這份作業完成後不需要繳交、評分。). But for practice,or quizzes or exams, please make sure that you have idea to solve the following questions by yourself. (這些題目是很重要的練習，小考、期中、期末考會有類似的題目。練習這些題目的過程中所逐漸形成我們對於所學到的知識的"看法"，會在考試中很有幫助。所以，請一定要了解如何解題與解題背後的想法是什麼。)

Please prove the following propositions are true. (請證明下列命題爲眞。)

**Section 8.1.**
8.6 (ii) Theorem (on page 288)
8.7 Remark (on page 288)